Bleeding, Reversal, HIT and DIC

Anticoagulant reversal, the 4Ts and HIT care, DIC and TTP for critical care exams

REVERSING ANTICOAGULANTS IN MAJOR BLEEDING

DRUG	REVERSE WITH	NOTES
Warfarin	4F-PCC plus vitamin K 10 mg IV (slow)	PCC 25–50 units/kg by INR; FFP only if no PCC
Dabigatran	Idarucizumab 5 g IV	Also removed by dialysis
Apixaban, rivaroxaban	Andexanet alfa where available, or 4F-PCC 50 units/kg	Time of last dose matters
Unfractionated heparin	Protamine 1 mg per 100 units given in last 2–3 h (max 50 mg)	Fast; watch for hypotension, anaphylaxis
Enoxaparin	Protamine 1 mg per 1 mg enoxaparin within 8 h	Only partial (about 60%)
Alteplase, tenecteplase	Cryoprecipitate; tranexamic or aminocaproic acid	Check fibrinogen
Antiplatelets	Platelets only before neurosurgery; consider desmopressin	Platelets worsened outcomes in ICH (PATCH)

Typical adult values for exam study. Follow your institution's protocols.

EXAM TIP Exams pair warfarin with PCC plus vitamin K (PCC is fast, vitamin K keeps the INR down) and dabigatran with idarucizumab.

HEPARIN-INDUCED THROMBOCYTOPENIA (HIT)

An antibody to PF4–heparin activates platelets: platelets fall and clots form. Thrombosis, not bleeding, is the danger.

Score 0–3 low (HIT unlikely, no testing); 4–5 intermediate; 6–8 high.

4TS ITEM	2 POINTS
Thrombocytopenia	Fall over 50% and nadir 20,000 or more
Timing	Fall on day 5–10, or day 1 with heparin in the past 30 days
Thrombosis	New clot, skin necrosis or reaction to a heparin bolus
Other cause	None apparent

- Intermediate or high score: stop ALL heparin (flushes, locks, LMWH, coated catheters)
- Start a non-heparin anticoagulant at once: argatroban (liver-cleared; good in kidney failure) or bivalirudin; fondaparinux or a DOAC when stable
- Send a PF4–heparin immunoassay; a functional test (serotonin release assay) confirms
- No routine platelet transfusion
- No warfarin until platelets recover (150,000 or more); then overlap 5 days or more

SAFETY Never use heparin flushes or heparin-coated lines after HIT. Document it as an allergy.

DIC AND TTP

	DIC	TTP
Cause	Sepsis, trauma, obstetric, malignancy	ADAMTS13 deficiency
Platelets	Low	Very low
PT and aPTT	Prolonged	Usually normal
Fibrinogen	Low (late)	Normal
Smear	May show schistocytes	Schistocytes (hemolysis)
Treatment	Treat the cause; replace if bleeding	Urgent plasma exchange; avoid platelets unless life-threatening bleeding

- DIC with bleeding: platelets if under 50,000; cryoprecipitate if fibrinogen under 150 mg/dL; plasma for a long PT/aPTT
- ISTH overt DIC score 5 or more: platelets, D-dimer, PT and fibrinogen

EXAM TIP Long PT/aPTT and low fibrinogen point to DIC; normal clotting times with hemolysis and neuro or kidney signs point to TTP.

Sources (6) are listed with links at icubasics.com/sheets/coagulation.html. For education and exam preparation only. Not medical advice; follow current guidelines and your institution's protocols. ICU Basics is an independent study aid not affiliated with any certifying body.